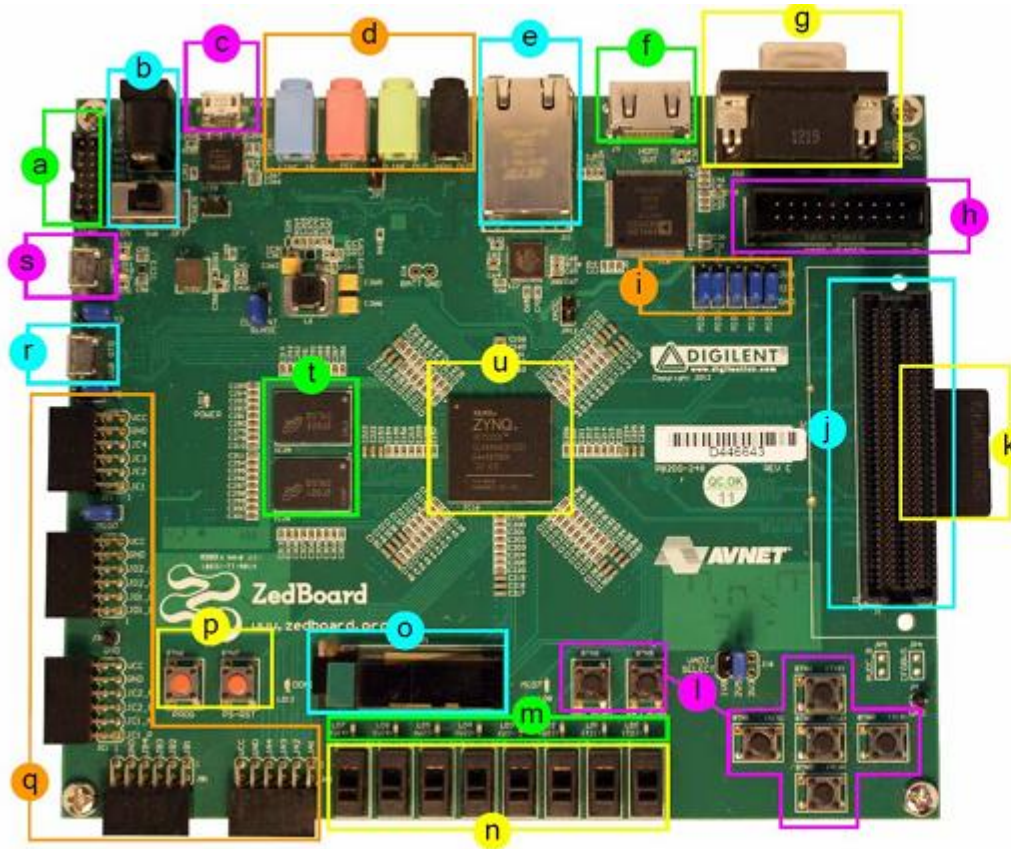
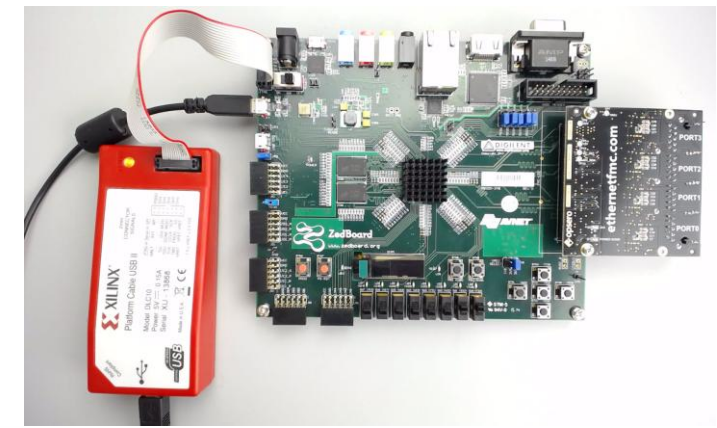


Zedboard in Detail

CSG553: Reconfigurable Computing

Meetha V Shenoy

ZedBoard Main Components

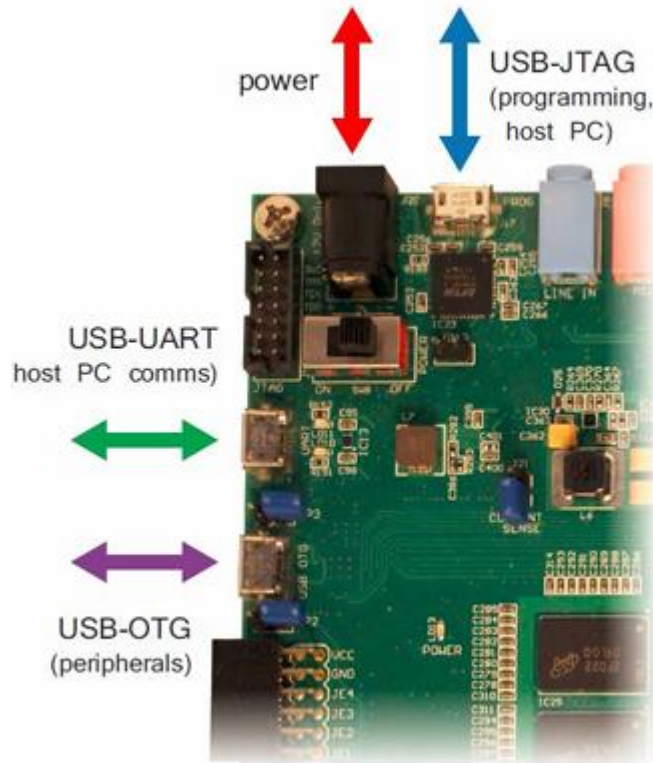


- a** Xilinx JTAG connector
- b** Power input and switch
- c** USB-JTAG (programming)
- d** Audio ports
- e** Ethernet port
- f** HDMI port (output)
- g** VGA port

GND	1	○	○	2	VREF
GND	3	○	○	4	TMS
GND	5	○	○	6	TCK
GND	7	○	○	8	TDO
GND	9	○	○	10	TDI
GND	11	○	○	12	---
GND	13	○	○	14	SRST

- h** XADC header port
- i** Configuration jumpers
- j** FMC connector
- k** SD card (underside)
- l** User push buttons
- m** LEDs
- n** Switches
- o** OLED display
- p** Prog & reset push buttons
- q** 5 x Pmod connector ports
- r** USB-OTG peripheral port
- s** USB-UART port
- t** DDR3 memory
- u** Zynq device (+ heatsink)

Main Connectors to be Used



By default, ZedBoard connects to the host PC for programming over USB-JTAG.

- An additional connection can be made over USB-UART.
 - If intending to facilitate simple board-PC communication using the Terminal application.
- Note that there is also a third micro-USB port for connecting USB peripherals (USB-OTG)

Zynq-7000 SoC

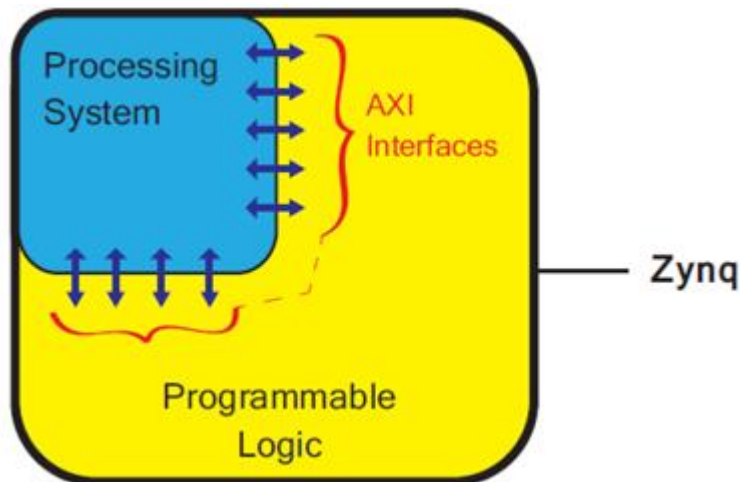
- Zynq comprises two main parts: –
 - Programmable Logic (PL): Artix-7 FPGA technology
 - Processing System (PS): a dual-core ARM Cortex-A9 processor

Spartan – very low cost, lower performance.

Artix – low power, medium performance.

Kintex – higher performance, higher cost.

Virtex – high-end, maximum performance and features.

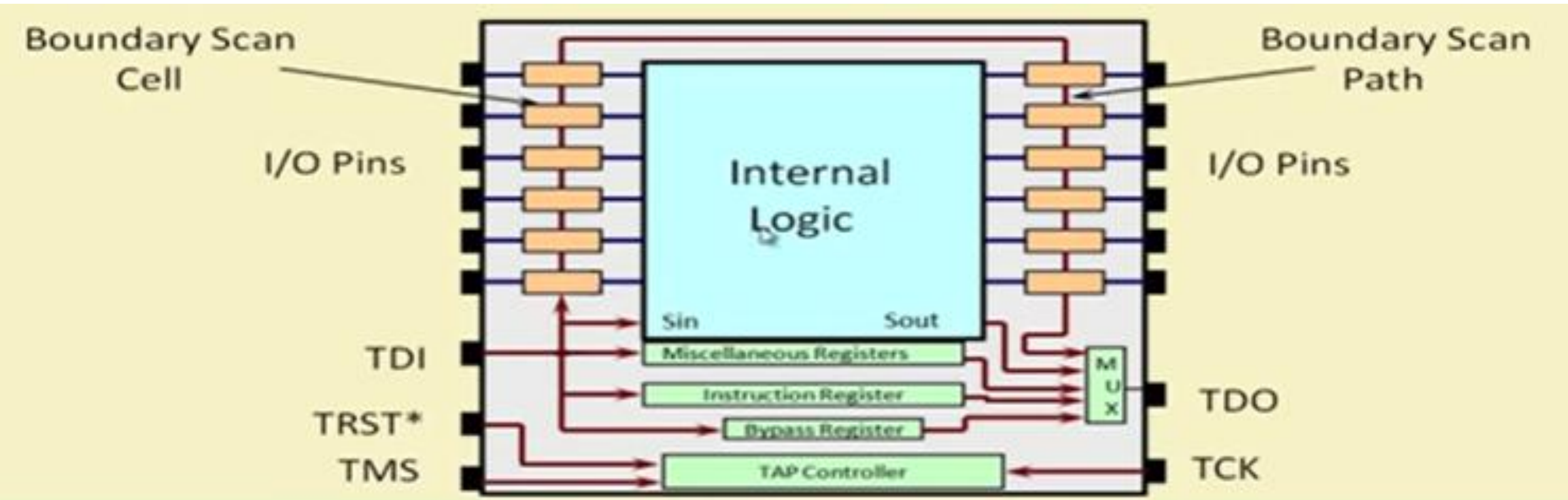
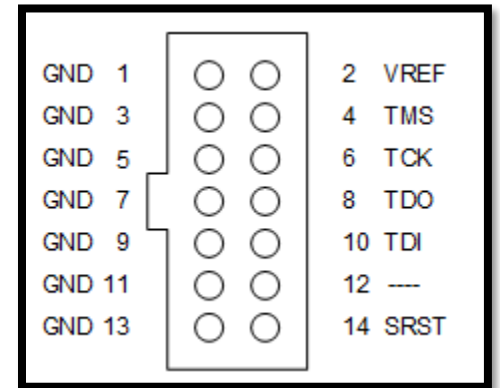


Advanced eXtensible Interface (AXI):– Links between the PL and PS are made using industry standard Advanced eXtensible Interface (AXI) connections.

Hard processors

Soft processors (Microblaze (32-bit RISC) in FPGA)

1149.1 Wrapper architecture

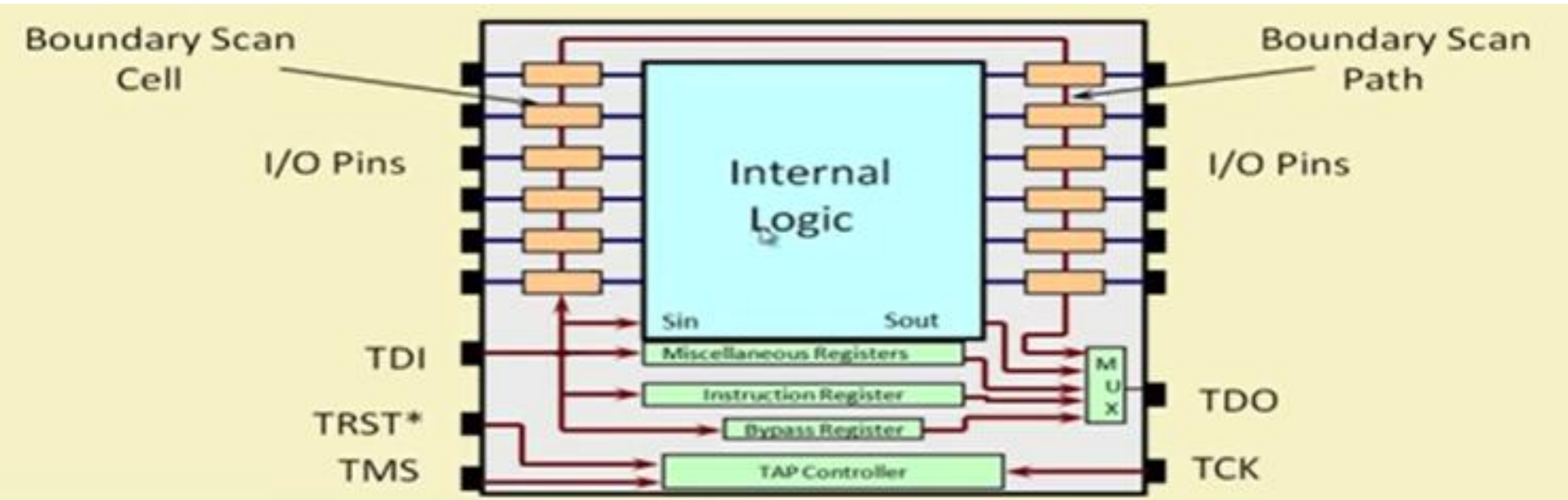
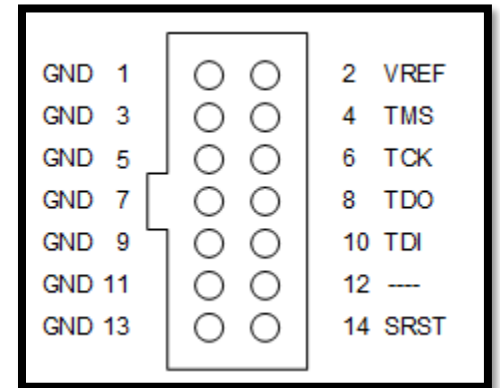


TAP signals

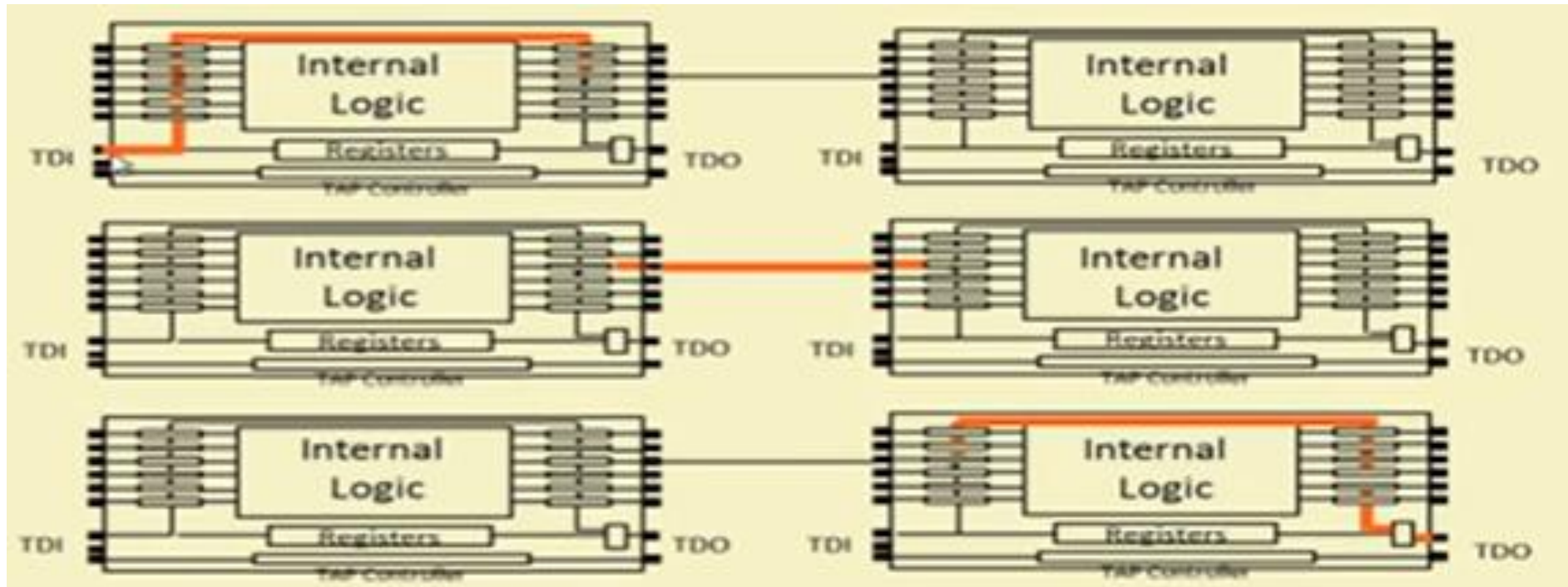
Mandatory JTAG Pins (4 or 5 signals)

- 1.TCK (Test Clock)** – Provides the clock for the boundary-scan shift register.
- 2.TMS (Test Mode Select)** – Controls the state machine of the TAP controller.
- 3.TDI (Test Data In)** – Serial input data for shifting through scan chains.
- 4.TDO (Test Data Out)** – Serial output from the scan chain.
- 5.TRST (optional Test Reset)** – Asynchronous reset for TAP controller.

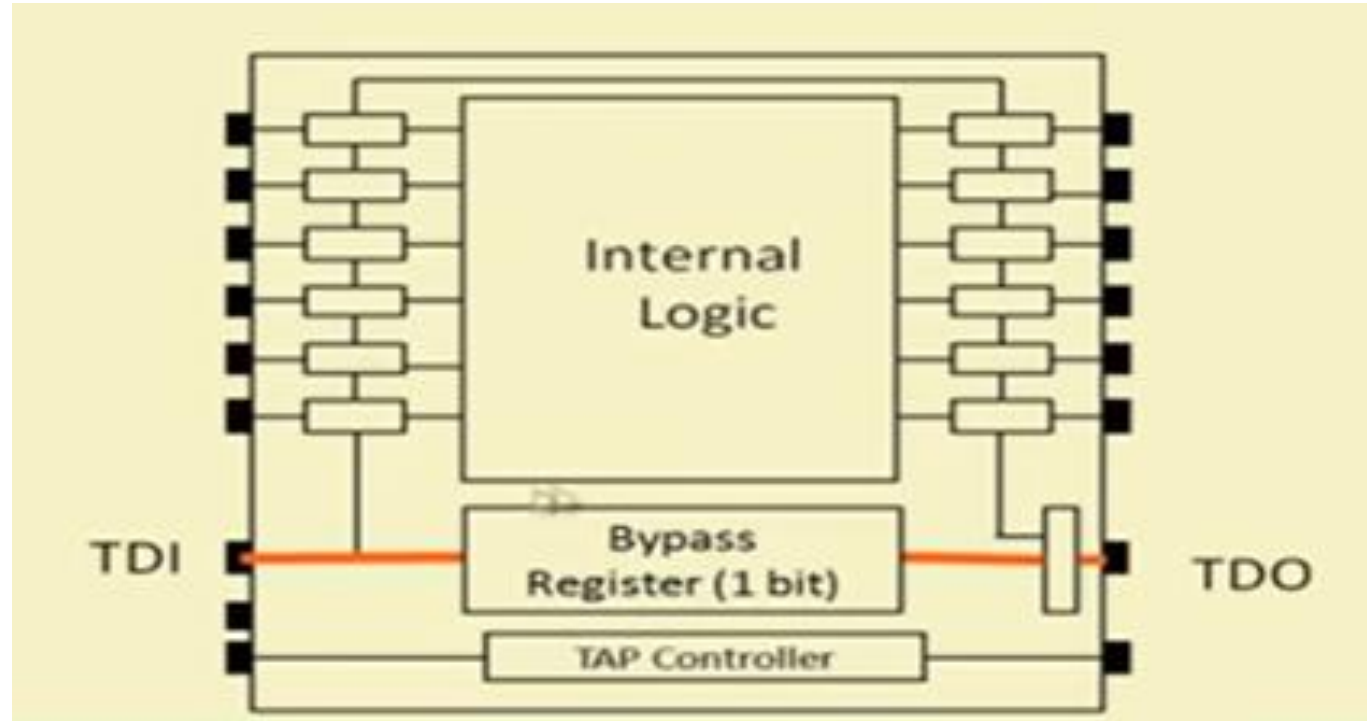
1149.1 Wrapper architecture



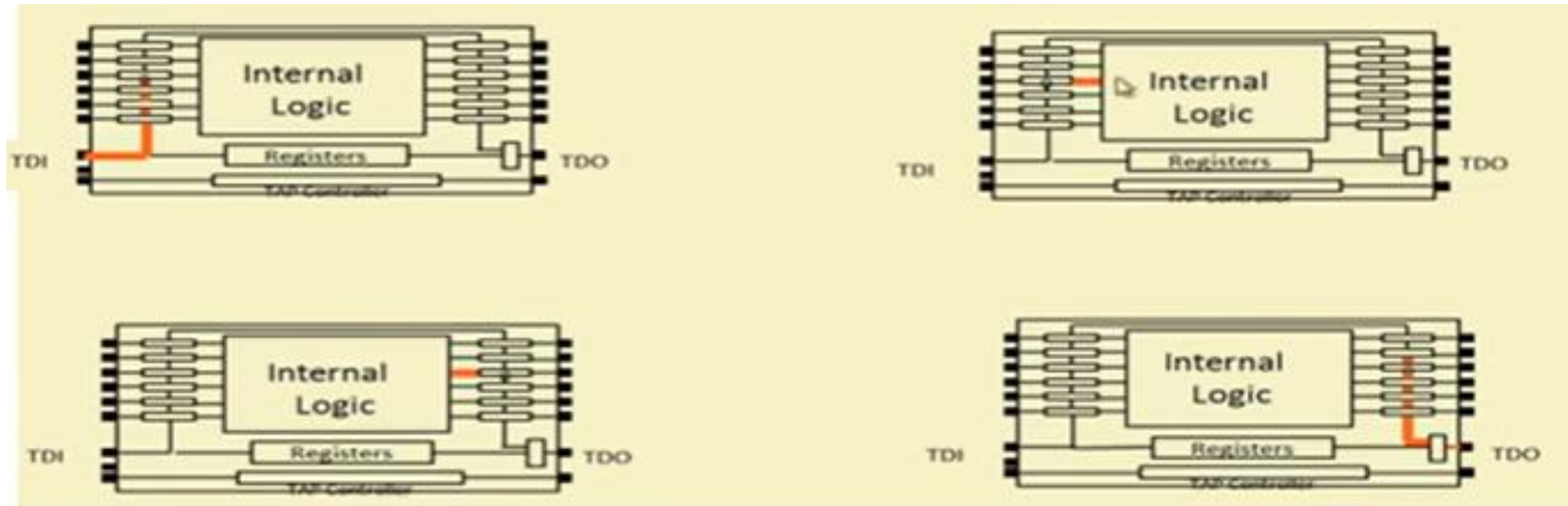
EXTEST MODE



BYPASS

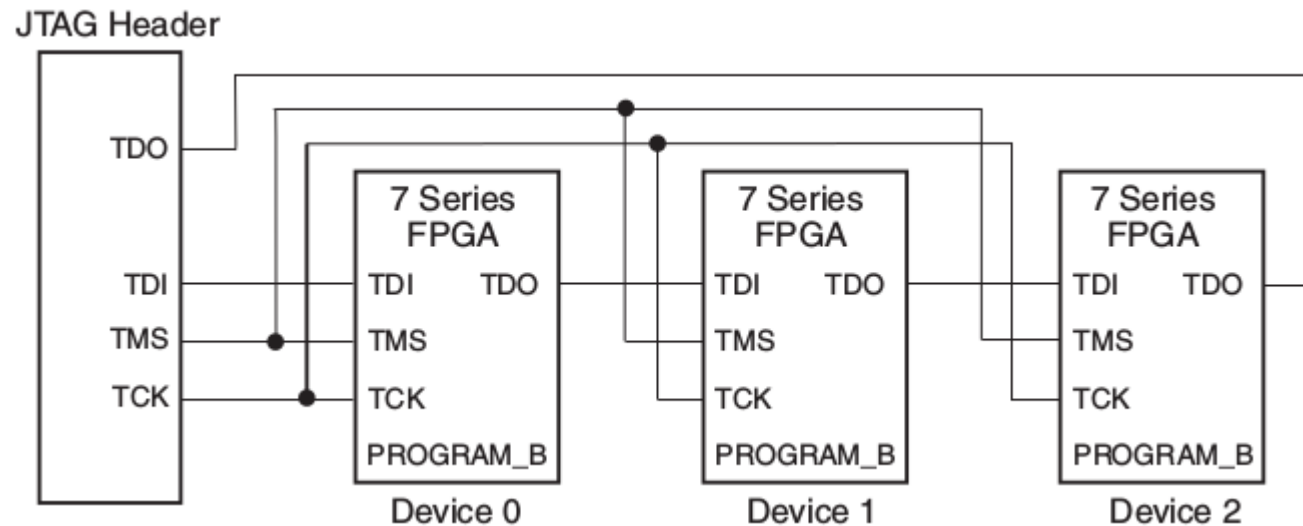


INTEST



JTAG

- JTAG = Joint Test Action Group, formulated hardware testing standard
→ IEEE 1149.1 (Boundary scan standard)



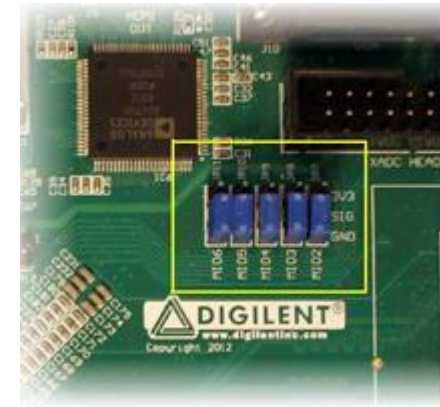
Programming the ZedBoard

- ZedBoard can be programmed in four different ways:
 - **USB-JTAG:** This is the default and most straightforward method of programming the ZedBoard, given that it can be done directly over the USB-micro-USB cable supplied in the ZedBoard kit.
 - **Traditional JTAG:** A Xilinx JTAG connector is available on the board and may be used in place of the USB-JTAG connection, if desired. This will require a different type of cable or a Digilent USB-JTAG programming cable.
 - **SPI flash memory:** The flash memory on the board is non-volatile and therefore can be used to store configuration data which persists when the board is powered off. Using this method removes the requirement for a wired connection to program the Zynq device.
 - **SD card:** There is an SD slot on the underside of the ZedBoard. This facility can be used to program the Zynq with files stored on the SD card, thus requiring no wired connections

Programming the ZedBoard

- The ZedBoard user specifies the method of booting / programming via a set of jumper pins.

	MIO[6]	MIO[5]	MIO[4]	MIO[3]	MIO[2]
In Xilinx Technical Reference Manual...	Boot_Mode[4]	Boot_Mode[0]	Boot_Mode[2]	Boot_Mode[1]	Boot_Mode[3]
<i>JTAG Mode</i>					
Cascaded JTAG ^a	-	-	-	-	0
Independent JTAG	-	-	-	-	1
<i>Boot Device</i>					
JTAG	-	0	0	0	-
Quad-SPI (flash)	-	1	0	0	-
SD Card ^a	-	1	1	0	-
<i>PLL Mode</i>					
PLL Used ^a	0	-	-	-	-
PLL Bypassed	1	-	-	-	-



Cascaded: A single JTAG connection is used to interface to the debug access ports in both the PS and PL